

COM410 Programming in Practice

A1. Object Oriented Concepts

CHALLENGE

Pet Hierarchy

- i. In a new Java project called Pets, define a new class `Pet` with instance variables `name` and `age`. Provide a suitable constructor and accessor/mutator methods for the class.
- ii. Now, define two new classes `Cat` and `Dog` as subclasses of `Pet`. Each sub-class has an additional instance variable `breed` that can store the particular type of cat (e.g. Persian, Tabby, etc.) or dog (e.g. Spaniel, terrier, etc.).
- iii. Each sub-class should contain a method called `speak()` that returns a typical animal noise, plus a description of the animal such as
"Miaow! I am Pixel, a 4 year old tabby", or
"Woof! I am Rex, a 9 year old terrier".
- iv. Provide a further class `PetTest` that implements a `main()` method to test your new class hierarchy.

Please upload a screenshot of your solution running to the A1 Challenge submission link on Blackboard, by the deadline shown there.